

SQL Window Functions – Quick Reference Guide

1. Introduction

Window Functions perform calculations across a set of rows related to the current row without reducing the number of rows returned.

GROUP BY	Window Functions
Many Rows → Few Rows	Many Rows → Many Rows
Returns Summary	Keeps Every Row
Aggregation	Analytics + Aggregation
Used for Reports	Used for Dashboards & Analytics

2. Window Function Syntax

```
SELECT columns,  
FUNCTION() OVER(  
    PARTITION BY column  
    ORDER BY column  
    ROWS/RANGE ...  
)  
FROM table;
```

OVER () – Defines the window.

PARTITION BY – Creates logical groups.

ORDER BY – Defines calculation order.

ROWS/RANGE – Defines the frame.

3. Ranking Functions

Function	Purpose	Business Use Case	Remember
ROW_NUMBER()	Unique numbering	Top-N, Deduplication	No ties
RANK()	Ranking	Leaderboards	Skips ranks after ties
DENSE_RANK()	Ranking	Prize tiers	No skipped ranks
NTILE()	Buckets	Quartiles, Salary Bands	Creates groups

4. Analytical Functions

Function	Purpose	Business Use Case	Remember
LAG()	Previous row	Month-over-month sales	Looks backward
LEAD()	Next row	Forecasting	Looks forward
FIRST_VALUE()	First value	First purchase	Returns value
LAST_VALUE()	Last value	Latest transaction	Frame matters
NTH_VALUE()	Nth value	Nth ranked record	Generalized value

5. Aggregate Window Functions

Function	Purpose	Business Use Case	Remember
SUM() OVER	Running total	Revenue dashboards	Keeps detail rows
AVG() OVER	Running average	Moving averages	Rolling analytics
COUNT() OVER	Count rows	Orders per region	No grouping
MIN() OVER	Running minimum	Lowest price	Running minimum
MAX() OVER	Running maximum	Highest revenue	Running maximum

6. Quick Comparisons

- **ROW_NUMBER vs RANK** → Unique numbers vs ties with skipped ranks
- **RANK vs DENSE_RANK** → DENSE_RANK never skips ranks
- **LAG vs LEAD** → Previous vs Next row
- **FIRST_VALUE vs LAST_VALUE** → First vs Last value
- **SUM () OVER vs GROUP BY SUM** → Detail rows retained vs collapsed
- **PARTITION BY vs GROUP BY** → Groups without reducing rows

7. Business Requirement → Function

Requirement	Function
Running Total	SUM() OVER
Running Average	AVG() OVER
Highest Salary per Department	MAX() OVER(PARTITION BY...)
Employee Ranking	RANK()/DENSE_RANK()
Unique Sequence	ROW_NUMBER()
Previous Record	LAG()
Next Record	LEAD()
Top N Per Group	ROW_NUMBER()+WHERE
Quartiles	NTILE()
First Purchase	FIRST_VALUE()
Latest Transaction	LAST_VALUE()

8. Interview Revision Sheet

- Window Functions never reduce rows.
- OVER() converts aggregate into window calculation.
- PARTITION BY creates logical groups.
- ORDER BY controls calculation order.
- ROW_NUMBER never repeats.
- RANK skips ranks after ties.
- DENSE_RANK does not skip ranks.
- LAG reads previous row.
- LEAD reads next row.
- Most common use: Reporting, Dashboards, Analytics.

Execution Order

FROM → WHERE → GROUP BY → HAVING → WINDOW FUNCTIONS → SELECT → ORDER BY

Memory Trick

GROUP BY = Summarize |
WINDOW = Analyze |
PARTITION = Divide |
ORDER BY = Arrange |
OVER () = Keep Rows